Hyperspectral and Multispectral Image Fusion Via Self-Supervised Loss and Separable Loss

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***Abstract*— Fusion of hyperspectral images (HSIs) with low-spatial and high-spectral resolution and multispectral images (MSIs) with high-spatial and low-spectral resolution is an important method to improve spatial resolution. The existing deep-learning-based image fusion technologies usually neglect the ability of neural networks to understand differential features. In addition, the loss constraints do not stem from the physical characteristics of the hyperspectral (HS) imaging sensors. We propose the self-supervised loss and the spatially and spectrally separable loss, respectively: 1) the self-supervised loss: different from the previous way of directly stacking the upsampled HSIs and MSIs as input, we expect the potentially processed HSIs to ensure not only the integrity of HSI information but also the most reasonable balance between overall spatial and spectral features. First, the preinterpolated HSIs are decomposed into subspaces as self-supervised labels. Then, a network is designed to learn subspace information and obtain the most discriminative features and 2) the separable loss: according to the physical characteristics of HSIs, the pixel-based mean square error loss is first divided into the domain loss and spectral domain loss, and then the similarity score of the images is calculated and used to construct the weighting coefficients of the two domain losses. Finally, the separable loss is jointly expressed by the weights. Experiments on public benchmark datasets indicate that the self-supervised loss and separable loss can improve the fusion performance.**

***Index Terms*— Discriminative samples' generation, hyperspectral (HS) and multispectral image (MSI) fusion, self-supervised loss, separable loss.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain not only spatial information but also abundant spectral information contributing to distinguish different materials in the scene. HSIs are used in various fields universally, such as medical field and remote sensing field including target detection [1], land cover classification [2], and so on, which enhance our capabilities of monitoring natural resources, the ecosystem, biodiversity, and dealing with disasters. However, limited by the inherent properties of the optical imaging system, the trade-off of resolution and signal-to-noise ratio (SNR) is inevitable. To acquire an HSI with high spatial resolution, the image fusion technology becomes an important means.

Compared with HSIs, multispectral images (MSIs) have fewer spectral bands but higher spatial resolution, and the spatial resolution of HSI is increased using spatial information of MSI or panchromatic (PAN). However, the hand-crafted-feature-based fusion methods have limited representational ability, and the premises of these methods are prior assumptions including endmembers [3], the sparsity prior in [4], [5], and others [6], which can lead to quality degeneration under the condition of unreasonable assumptions.

Deep learning can model the complex nonlinear relationship relying on data-driven [7] without any assumption or prior knowledge. A super-resolution convolutional neural network (SRCNN) [8] converts the image restoration problem into the mapping relationship between the input image and the desired output for the first time. Different from the single-image super-resolution, there are two input images in the image fusion of low-resolution HSI (LR-HSI) and high-resolution MSI (HR-MSI). Therefore, inspired by a single-image restoration strategy, the HSI and MSI fusion is transformed into a multi-input to single-output mapping relationship. To obtain HR-HSIs, the network structures [9], [10], [11], [12] of single-image restoration are optimized and used for the fusion of HSI and MSI. However, more methods are concerned with the performance improvement caused by the network structures, while ignoring the ability of neural networks to understand differential features, that is, the features extracted in different ways have differences for subsequent fusion. In addition, for the loss function, these methods [13], [14], [15], [16] in HSI restoration simply apply the same loss functions as red, green, blue (RGB) image restoration and directly use mean square error (MSE) or its related loss function to constrain the training process. In terms of these methods, HSI is only an image that does not need to distinguish between spatial and spectral information, as long as the error of pixel values is as small as possible. However, compared with RGB, HSI has rich spectral information, so the loss functions based on MSE ignore the physical meaning of HSI.

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In this article, we propose the self-supervised loss (Suv) and separable loss (Lss) function for HSI and MSI fusion. It is different from the way that the original HSIs or reduced dimensional HSIs are directly input into the network, and the self-supervised loss (Suv) aims to improve the understanding ability of the neural networks by analyzing feature fusion and extract the most discriminative features for subsequent fusion. The most discriminative features refer to the features to achieve optimal fusion in this article. First, the supervised learning labels are produced using principal component analysis (PCA), and we design a dimensionality reduction network that reduces data redundancy and noise and balances spectral and spatial data. Then the learned feature maps of HSI are stacked with structural feature maps by convolutional neural network (CNN) to produce the most discriminative features for the fusion task. The Lss is based on the physical properties of HSI, and we divide the pixel-based MSE loss into spatial loss and spectral loss and calculate the similarity score of the HSI to construct the weight coefficients of the two losses. Finally, the separable loss is jointly expressed by two domain losses and weights.

II. Related Work

Recent researches obtain HSI using spectral resolution enhancement. Some methods based on dedicated hardware equipment acquire them by RGB cameras with active light [17] and tunable narrowband filters [18]. Nguyen *et al*. [19] reconstruct an HSI from an RGB image based on a known spectral response function and a proposed radial basis function network. The degradation adaptive learning [20] is also concerned. The literature [21], [22] recovers hyperspectral (HS) pixels using priors. In [23], [24], the spectral resolution enhancement is regarded as a mapping between RGB images and HSIs. The articles [25], [26], [27] use a generative model to learn mapping. Yan *et al*. [23] propose a downsampling and upsampling CNN, which reconstructs the spectral information by mining the local and nonlocal image information. The main way to obtain HSI through spectral super-resolution is to preserve the spatial resolution and improve the spectral resolution, and most of the approaches cope with limited spectral range.

HSI and MSI fusion has received more and more attention, and CNN is widely used to solve the classic image fusion problems. The existing CNN-based methods can be split into two categories: input-level fusion and feature-level fusion.

In the input-level fusion, more articles [28], [29], [30] regard the multiple-input and single-output image fusion as a single-input and single-output restoration problem. For example, Palsson *et al*. [31] and Masi *et al*. [30] stack the LR-HSI with HR-MSI along the spectral channels and then fed it into a deep CNN. The way of directly stacking data neglects a general balance between the spatial and spectral feature maps. Han *et al*. [32] add the shortcut connection to extract more spatial details. For networks, more discriminative features and enough data are vital. The article [16] strengthens the discrimination of samples via reducing the spectral dimension gap with CNN. HSIs are highly redundant. Band selection and dimension reduction are implemented [33], [34] refraining from the problem of high dimensionality of HSI. It can be seen that PCA not only contains useful information of the original image but also removes part of the noise. Similarly, the subspace-based methods [33], [34] and low-rank [35] technologies greatly improve the efficiency of the models and make the methods more robust to noise. The existing methods [31] based on deep learning often use subspace technologies to reduce redundancy of HSI. Although the method retains most of the information of HSI, it does not focus on whether the information distribution in the transformed low-dimensional space is the optimal for the fusion task.

In the feature-level fusion, more HSI and MSI fusion strategies are proposed, and various structures continue to emerge. Wei *et al*. [9] propose a deep residual pansharpening neural network (DRPNN), and Zheng *et al*. [11] adopt a dual-attention residual network (DARN) for HS pansharpening, and the residual network (ResNet) is taken to extract high-level features enhancing image details. A 3-D generative adversarial network (3-D GAN) [10] is built to implement HS pansharpening. It achieves spectral–spatial joint learning instead of focusing on spatial pixels. Similarly, many related technologies [31], [36] have been proposed. Based on this observation that HSI has rich spectral information and considering the spectral distortion caused by direct stacking two kinds of images, the articles [37], [38] establish a dual-channel CNN extracting spatial and spectral features, respectively. Other works [39], [40] draw lessons from the traditional detail-injected mechanism that spatial details from the HR-MSI are injected to the LR-HSI to keep better spectral fidelity. The researches show that the learning ability of CNN is different for high-frequency details and smooth images, and each channel of HSI has a different influence on the fused image. Numerous excellent works [41], [42] have shown that the discriminative ability of CNN can be enhanced through the attention mechanism or balancing the proportion of spatial and spectral features. The mentioned methods transfer the characteristics of the images to the network architectures, and they just stack or extract the input data separately instead of analyzing data to get the most discriminative features.

The methods based on deep learning mainly study the mapping relationship between the input images and an output image for image fusion, and it is necessary to construct a suitable loss function for training a robust and high-performance model. The general loss functions are selected such as root-mean-square error (RMSE), and these methods [14], [15] adopt them the same as the RGB image. Considering that HSIs contain both spatial and spectral information, Yang *et al*. [38] propose spectrum learning by the deep learning network, Xie *et al*. [10] use spectral angle mapper (SAM) as a loss to constraint training, and He *et al*. [43] add the constraints of spectral information based on the spectral angle distance to decrease the spectrum distortion.

III. Proposed Model

It is an important means to improve the image resolution using the HS and multispectral fusion technology, and this section discusses the proposed technologies in detail. As seen in Fig. 1, we show the flow graph of the image fusion. The

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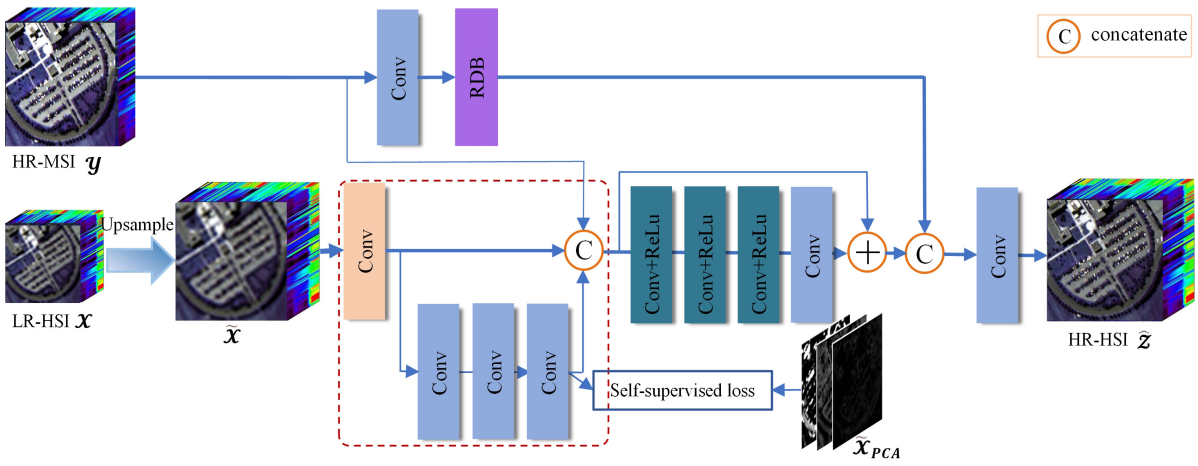

Fig. 1. Flow graph of the image fusion; "Conv" donates the convolutional layer, "Conv+Relu" represents the convolutional layer with activation function ReLu, and "RDB" is the residual dense block. Convs with different colors have convolution kernels with different sizes.

original LR-HSI is represented as $\mathcal{X} \in \mathbb{R}^{h' \times w' \times c}$, where h', w', and c are the height, width, and the number of spectral bands, respectively. $\mathcal{Y} \in \mathbb{R}^{h \times w \times c'}$ is the HR-MSI, and h, w, and c' are the height, width, and the number of spectral bands of HR-MSI, respectively, in which the spatial dimension $h' < h$, $w' < w$, and the spectral dimension $c' < c$. $\mathcal{Z} \in \mathbb{R}^{h \times w \times c}$ is the desired HR-HSI. $\tilde{\mathcal{X}} \in \mathbb{R}^{h \times w \times c}$ represents the up-interpolated LR-HSI by the bicubic interpolation method, where the image fusion scale factor $s = h/h'$. $\hat{\mathcal{Z}} \in \mathbb{R}^{h \times w \times c}$ denotes the fused HR-his, and the spatially and spectrally downsampled versions of $\mathcal{Z}$ are the LR-HSI and HR-MSI, respectively. The image fusion aims at estimating $\mathcal{Z}$ by combining $\mathcal{X}$ with $\mathcal{Y}$. To achieve the goal, a two-branch CNN is designed. The top MSI branch provides the main spatial detail information. Considering that the residual dense block (RDB) can continuously connect the front and rear features, we adopt an RDB module to make the MSI provide sufficient spatial detail. The bottom branch is used to maintain the image spectral information while enhancing the spatial detail. However, this branch contains both HSI and MSI image information. RDB can cause the problem of slow convergence of spectral information by interleaving features. Therefore, the general CNN structure is adopted. Based on the network structure, our proposed technique for image fusion mainly consists of two parts: one learns low-dimensional discriminative samples from preinterpolated HSI by the self-supervised loss, and the other supervises the training process based on the 3-D characteristics of HSI.

A. Discriminative Sample Generation Based on Self-Supervised Learning

We aim at learning the most discriminative features and reducing HSI dimensions at the same time. Although the traditional dimension reduction methods can retain most of the information of the image, it lacks discrimination. We use the way that the low-dimensional HSI is learned from the original HSI. The dimension of the original HSI is reduced, and the reduced dimensional image is used to generate weakened labels that contain almost all the useful information of HSI. The acquisition process of labels is as follows.

The up-interpolated LR-HSI $\tilde{\mathcal{X}} \in \mathbb{R}^{h \times w \times c}$ is represented as a matrix $\tilde{\mathbf{X}} = (\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_c) \in \mathbb{R}^{N \times c}$, where $N = w * h$ represents the number of pixels. We first perform singular value decomposition (SVD) on the deaveraged image $\tilde{\mathbf{X}} \in \mathbb{R}^{N \times c}$

$$\tilde{\mathbf{X}} = \mathbf{U} \sum \mathbf{V}^T \tag{1}$$

where $\mathbf{U} \in \mathbb{R}^{N \times N}$ and $\mathbf{V} \in \mathbb{R}^{c \times c}$ represent a left singular matrix and a right singular matrix, respectively, and $\sum \in \mathbb{R}^{N \times c}$ consists of singular values. Then image compression along the column is executed via regarding the right singular matrix as the transformation matrix, and the image $\tilde{\mathbf{X}}_{\mathrm{PCA}} = (\tilde{x}_{\mathrm{PCA1}}, \tilde{x}_{\mathrm{PCA2}}, \ldots, \tilde{x}_{\mathrm{PCAk}}) \in \mathbb{R}^{N \times k}$ after being dimensioned is as follows:

$$\tilde{\mathbf{X}}_{\mathrm{PCA}} = \tilde{\mathbf{X}} \mathbf{V}_k \tag{2}$$

$\mathbf{V}_k = (v_1, v_2, \ldots, v_k) \in \mathbb{R}^{c \times k}$ is a transformation matrix from the first k columns of the matrix $\mathbf{V}$ and $k \ll c$. We therefore can acquire the 3-D tensor $\tilde{\mathcal{X}}_{\mathrm{PCA}} \in \mathbb{R}^{h \times w \times k}$, where k is the spectral band number.

To extract more discriminative features in a supervised way, we design a simple three-layer CNN seen from the red dashed box in Fig. 1. Since the dimension reduction process is a linear operation, only simple convolution layers are used for feature learning. The purpose of the bottom branch in the box is not to learn the same feature maps as $\tilde{\mathcal{X}}_{\mathrm{PCA}}$, but to extract more discriminative feature maps and reduce the dimension of the image $\tilde{\mathbf{X}}$. Consequently, this requires weakening supervision, and we give a weight α to the self-supervised loss in accordance with the sample characteristics

$$\mathcal{L}_{\mathrm{suv}} = \frac{\alpha}{\mathrm{hwk}} \sum_{i}^{h} \sum_{j}^{w} \sum_{p}^{k} \left\| \hat{\mathcal{X}}_{\mathrm{PCA}}(i, j, p) - \tilde{\mathcal{X}}_{\mathrm{PCA}}(i, j, p) \right\|_F^2 \tag{3}$$

where $\hat{\mathcal{X}}_{\mathrm{PCA}} \in \mathbb{R}^{h \times w \times k}$ is an estimated low-dimensional discriminative image. i, j, p represents the indexes of image

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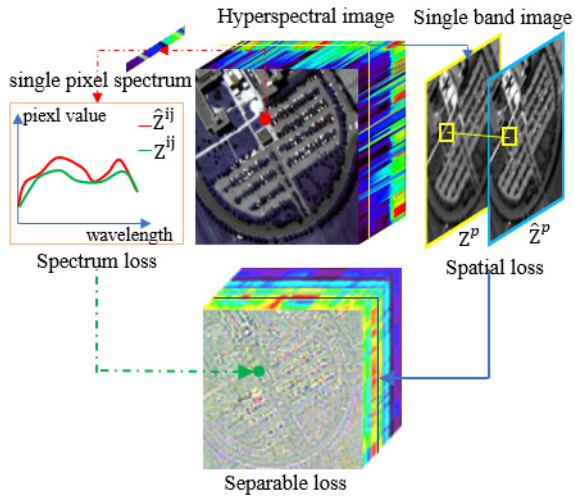

Fig. 2. Schematic of the separative loss.

dimensions. Compared with $\tilde{\mathcal{X}}$, $\hat{\mathcal{X}}_{\text{PCA}}$ is more discriminative and less redundant. Since $\tilde{\mathcal{X}}$ and $\mathcal{Y}$ are quite different in spectral and spatial resolution and $\tilde{\mathcal{X}}$ consists of a large number of spectral bands with low resolution, which increases the proportion of low-resolution images in the overall image. So, the generated samples based on self-supervised learning enhance the balance. In addition, the features $\hat{\mathcal{X}}_{\text{PCA}}$ are linearly independent, and the original information distribution has been changed. For retaining the original image structure, we use the conventional feature extraction strategy, as shown in the upper branch of the red dashed box. The normal CNN feature maps $\mathcal{X}_{\text{CNN}} \in \mathbb{R}^{h\times w\times d}$ with the original image structures are stacked with the features $\hat{\mathcal{X}}_{\text{PCA}}$ to form discriminative feature maps of HSI, where d represents the spectral band number, and $d < c$. Eventually, the input of the backbone network consists of stacked features $(\mathcal{X}_{\text{CNN}} \| \hat{\mathcal{X}}_{\text{PCA}} \| \mathcal{Y})$, and "||" denotes the concatenation in the third dimension.

B. Separative Loss Function

The loss function can be a good tool to reflect the gap between the model and the actual data, it is crucial to find the most reasonable loss function for HSI restoration. In this article, we have deeply thought about the characteristic of HSI containing both spatial and spectral information, which motivates us to restrict training from two aspects. As shown in Fig. 2, HSI contains hundreds of single bands in the spatial domain, and we expect that the loss between each band $\hat{\mathcal{Z}}(p)$ and the corresponding actual band $\mathcal{Z}(p)$ is the smallest. Therefore, we consider both spatial and spectral differences here and regard each pixel as a key point and calculate the similarity score

$$\mathcal{C} = E - \frac{\hat{\mathcal{Z}}^T \mathcal{Z}}{||\hat{\mathcal{Z}}||||\mathcal{Z}||} \tag{4}$$

where $w = h$, $\mathcal{C} \in \mathbb{R}^{h\times w\times c}$, and $E \in {}^{h\times w\times c}$ is a 3-D unit matrix. The spatial difference matrix is obtained by the similarity score to reduce the spatial difference

$$\mathbf{C}_{\text{spa}} = \sum_{i}^{h}\sum_{j}^{w} \mathcal{C}(i,j) \tag{5}$$

where $\mathbf{C}_{\text{spa}} \in \mathbb{R}^c$. For the total difference of each 2-D image in the spatial domain $\mathbf{C}_{\text{spa}}$, it is normalized as a spatial coefficient matrix spatial

$$\mathbf{w}_{\text{spa_norm}} = \frac{\mathbf{C}_{\text{spa}} - (\mathbf{C}_{\text{spa}})_{\min}}{(\mathbf{C}_{\text{spa}})_{\max} - (\mathbf{C}_{\text{spa}})_{\min}}. \tag{6}$$

Therefore, the spatial loss function can be expressed as

$$\mathcal{L}_{\text{spa}} = \frac{1}{c}\sum_{p}^{c}\sum_{j}^{w}\sum_{i}^{h} w_{\text{spa_norm}}(p)\left\|\hat{\mathcal{Z}}(i,j,p) - \mathcal{Z}(i,j,p)\right\|_F^2 \tag{7}$$

where $w_{\text{spa_norm}} \in \mathbb{R}^c$.

For spectrum loss, each spatial pixel $\mathcal{Z}(i,j)$ corresponds to a spectral curve $S(r), r \in R^N$ in the spectral domain and we expect the error of each curve to be minimized. In the same way as creating the spatial difference matrix, we use the similarity score matrix $\mathcal{C}$ to construct the spectral difference matrix

$$\mathbf{C}_{\text{spe}} = \sum_{p}^{c} \mathcal{C}(p) \tag{8}$$

where $\mathbf{C}_{\text{spe}} \in \mathbb{R}^{h\times w}$. For spectral error $\mathbf{C}_{\text{spe}}$, it is normalized as a spectral difference matrix

$$\mathbf{w}_{\text{spe_norm}} = \frac{\mathbf{C}_{\text{spe}} - (\mathbf{C}_{\text{spe}})_{\min}}{(\mathbf{C}_{\text{spe}})_{\max} - (\mathbf{C}_{\text{spe}})_{\min}}. \tag{9}$$

The spectral loss is expressed as follows:

$$\mathcal{L}_{\text{spe}} = \frac{1}{wh}\sum_{i}^{h}\sum_{j}^{w}\sum_{p}^{c} w_{\text{spe_norm}}(i,j)\left\|\hat{\mathcal{Z}}(i,j,p) - \mathcal{Z}(i,j,p)\right\|_F^2 \tag{10}$$

where $w_{\text{spe_norm}} \in \mathbb{R}^{h\times w}$. Therefore, compared with the conventional way of using the RMSE to constrain the training process, we separate the loss using the characteristics of HSI as shown in (7) and (10).

In the training process, the total loss is

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{ss}} + \mathcal{L}_{\text{suv}} = \mathcal{L}_{\text{spa}} + \mathcal{L}_{\text{spe}} + \mathcal{L}_{\text{suv}}. \tag{11}$$

IV. Experiments Results

In this section, there are a real dataset and two simulated datasets to evaluate the effect of the proposed model.

A. Experiment Setting

1) Experimental Database: Dataset 1: The Pavia University HSI with a 1.3-m geometric resolution is taken by the reflective optics system imaging spectrometer (ROSIS) optical sensor, and the area is located at the urban area of the University of Pavia, Italy. There are 115 bands with a band length of 430–860 nm, of which 12 bands are removed and the other 103 bands are used in the experiments. The image size

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of each band is 610 × 340, of which the area 400 × 336 is used for training, and the size 176 × 296 is for the tests. The spectral reflectance response filter [44] is matrix-multiplied with the HSIs to generate simulated high-spatial-resolution and low-spectral-resolution MSIs. A 7 × 7 Gaussian kernel with a standard deviation of 2 is regarded as a spatial blur kernel, and it is used to convolve with the HSI, and then the downsampling operation is performed using a scale factor of 8 to acquire simulated low-spatial-resolution and high-spectral-resolution HSIs.

Dataset 2: The Chikusei dataset is captured by headwall visible and near-infrared series C (VNIR-C) imaging sensor, and it includes agricultural and urban areas in Chikusei, Ibaraki, Japan. Its spatial resolution is 2.5 m, and the wavelength range covered is 363–1018 nm. There are 128 bands, and the whole scene consists of 2517 × 2235 pixels, of which 1440 × 864 pixels HSI is for training, and the size of the test HSI is 248 × 264. To produce the corresponding LR-HSI and HR-MSI with eight bands, a spatial sampling technology similar to that used for the first dataset and a spectral reflectance response filter is applied, respectively.

Dataset 3: The real LR-HSI is obtained by the Hyperion sensor onboard the Earth Observing-1 satellite, and its spatial resolution is 30 m. The LR-HSI contains 220 spectral bands in 400–2500 nm, of which 131 bands have been removed due to low SNR, and the remaining ones are used in this experiment. The image size of LR-HSI is 2350 × 990, the area 219 × 273 is for training, and the area 120 × 171 is used for the tests. The Sentinel-2A satellite captures the real HR-MSI consisting of 13 spectral bands, of which four bands are for the fusion task, and its spatial resolution is 10 m. The size of the covered space area by HR-MSI is 7050 × 2970, and we use 657 × 819 HR-MSI in the training and the area 360 × 513 is used in this test. The Hyperion HSI and the Sentinel-2A MSI were taken over Lafayette, LA, USA, in October and November, 2015, respectively. According to the articles [38], the cropped subimages are regarded as study areas from the overlapped region of the Hyperion and Sentinel data.

2) Quantitative Metrics: To evaluate the effect of recovered HSIs fairly and effectively, six quantitative indicators are calculated in this experiments, including RMSE, peak signal-to-noise ratio (PSNR), SAM [45], relative dimensionless global error in synthesis (ERGAS) [46], universal image quality index (UIQI) [47], and structural similarity (SSIM) [48]. Among them, UIQI and SSIM are used to evaluate the spatial information of the image, SAM and ERGAS indicators are used to evaluate the spectral information of the image, and RMSE and PSNR can evaluate the overall quality of the image.

3) Compared Methods: For our comparison, we choose traditional models including LTTR [49], FUSE [44], HySure [50], LTMR [35], and OTD [51]. For the sake of fairness, few simple models based on CNN, RSIFNN [37], Hyperpnn [16], TFNet [52], and (PDCon)SSF-CNN [32], are introduced to verify the effectiveness of our model. Meanwhile, these models, PCA, PCA+CNN, CNN, Suv, CNN+Lss, and Suv+Lss, are used to verify the effectiveness of different loss functions, where PCA, PCA+CNN, CNN, and Suv represent different feature extraction methods for HSI in the red dotted box in Fig. 1.

TABLE I
PARAMETERS OF OUR NETWORK STRUCTURE

		Layer	Filter	Stride	Output Size
Input	MSI				12*12*m
	HSI				12*12*n
Feature extraction	MSI	Conv1+Relu	3*3/64	1	12*12*64
		RDB			12*12*n
	HSI	Conv2+Relu	3*3/64	1	12*12*64
		Conv3+Relu	1*1/80	1	12*12*80
		Conv4+Relu	3*3/10	1	12*12*10
		Conv5+Relu	3*3/10	1	12*12*10
Input_level fusion		Concat			12*12*10
Feature_level fusion		Conv6+Relu	3*3/118	1	12*12*118
		Conv7+Relu	3*3/118	1	12*12*118
		Conv8+Relu	3*3/118	1	12*12*118
		Conv9	1*1/118	1	12*12*118
		Add			12*12*118
		Concat			12*12*(118+64)
Restoration		Conv10	3*3/n	1	12*12*n

4) Experimental Details: The deep learning framework PyTorch is regarded as a tool achieving all the deep-learning-based fusion methods, the traditional methods are accomplished, and the performance indexes are computed using MATLAB R2018b in Windows 10 operating system.

In the experiment, the main framework adopts the simple CNN architecture with five layers in Fig. 1, and the network structure parameters are given in Table I on the Pavia University dataset, where the label images of self-supervised learning are composed of the first 10 principal components ($k = 10$) of the HSI processed by PCA. The optimizer is Adam [53]. In the training and validation, the initial learning rate is 3×10^{-4}, the patch size is designed as 12, and the batch sizes of 32 and 16 are used for the experiment. For training datasets Pavia University, Chikusei, and Earth Observing-1 Hyperion and Sentinel-2A, the epoch numbers are 80, 70, and 60, respectively, and the coefficient α of the self-supervised loss in Suv+lss is 0.008.

The parameter $k = 10$ is obtained by referring to the empirical results of the excellent article [35]. For the loss weight during training, we follow the general parameter adjustment methods based on deep learning. The parameter α ensures that several losses are of the same order of magnitude. It makes the loss $\mathcal{L}_{\text{suv}}$ slightly smaller than other losses when the trained model is in convergence. Therefore, the operation guarantees that the overall loss $\mathcal{L}_{\text{ss}}$ of the image is minimized in the case of extracting the most suitable features for fusion.

B. Comparison With Methods

1) Experiments on Pavia University Dataset: We demonstrate the validity of the proposed structure by comparing with or without the self-supervision loss and separable loss in this section. The captured area contains more small objects, and the image includes more details.

Table II shows the quantitative values of compared methods, and the traditional image fusion methods LTTR, FUSE, HySure, LTMR, and OTD can restore image details and keep good spectral information. Excellent algorithms, such

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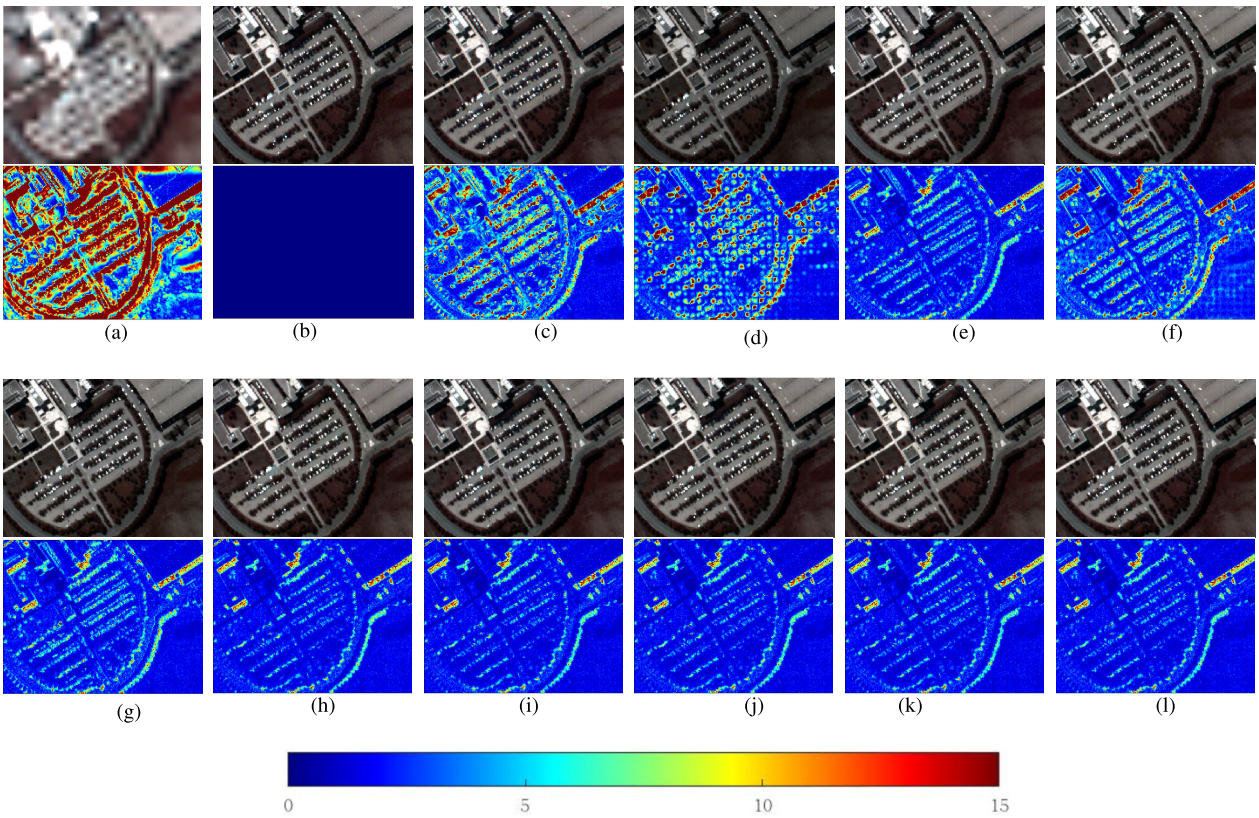

Fig. 3. False-color images (composed by bands 25, 15, and 10) are shown in the first and third rows, and SAM images are presented in the second and fourth rows. (a) LR-HSI. (b) Ground truth. (c) LTTR. (d) FUSE. (e) LTMR. (f) Hysure. (g) OTD. (h) RSIFNN. (i) Hyperpnn. (j) TFNet. (k) (PDCon)SSF-CNN. (l) Suv+Lss.

TABLE II

PERFORMANCE COMPARISON OF DIFFERENT FUSION METHODS ON A SUBSCENE OF THE PAVIA UNIVERSITY DATASET

	PSNR	SAM	ERGAS	SSIM	RMSE	UIQI
Reference	∞	0	0	1	0	1
LTTR [49]	36.2495	3.9865	1.2874	0.9622	3.9270	0.9674
FUSE [44]	36.3828	3.9767	1.3966	0.9656	3.8672	0.9649
HySure [50]	39.1532	3.3299	0.9525	0.9772	2.8111	0.9784
LTMR [35]	41.5787	2.4254	0.7631	0.9822	2.1262	0.9829
OTD [51]	41.0914	2.4852	0.8394	0.9818	2.3245	0.9829
RSIFNN [37]	40.0034	2.7564	0.9425	0.9797	2.5490	0.9795
RSIFNN+Lss	40.2305	2.7375	0.9122	0.9794	2.4832	0.9793
Hyperpnn [16]	40.9543	2.3926	0.8268	0.9824	2.2847	0.9826
Hyperpnn+Lss	41.2563	2.3624	0.8175	0.9826	2.2066	0.9828
TFNet [52]	41.5068	2.3684	0.7937	0.9828	2.1439	0.9832
TFNet+Lss	41.5632	2.3375	0.7961	0.9828	2.1300	0.9831
(PDCon)SSF-CNN [32]	42.0375	2.1918	0.7378	0.9846	2.0168	0.9849
(PDCon)SSF-CNN+Lss	42.1614	2.1676	0.7329	0.9847	1.9883	0.9851
Suv+Lss	**42.3018**	**2.1324**	**0.7192**	**0.9852**	**1.9564**	**0.9858**

as LTMR and OTD, show good performance. In Table II, in general, the fusion strategies based on deep learning, such as TFNet and (PDCon)SSF-CNN, perform well due to the strong ability of feature expression. Compared with the above methods, the quantitative performance evaluation shows the effectiveness of our Suv+Lss model. Here, RSIFNN+Lss, Hyperpnn+Lss, TFNet+Lss, and (PDCon)SSF-CNN+Lss denote the models after replacing MSE loss with our Lss loss, and the experimental results verify that our Lss loss is effective for image fusion. Therefore, it is key to enhance the ability of the network to extract suitable features for image fusion, that is, the ability of the network to understand differential features.

To observe intuitively the fusion effect of related methods in Fig. 3, the false-color images are constructed by bands 25, 15, and 10. The SAM images are captured by all the bands for different testing methods, where blue to red indicates that the SAM value increases gradually. On the whole, the smooth region of the images is restored better including little spectral distortion. Compared with interpolated LR-HSI, the images recovered by these methods obviously contain more details and it can be seen that the red area is significantly reduced in SAM images. However, compared with the ground truth, especially the spatial detail information and the spectrum at the detail need to be enhanced. It can be seen from Fig. 3 in the SAM image of HySure, LTTR, and FUSE that there are more red areas in image detail. Others show similar SAM images with spectrum differences in few areas, which is hard to discriminate with the naked eye.

With the aim of fully verifying the effectiveness of self-supervised loss and separable loss, we conduct experiments again based on the network structure in this article. In Table III, PCA, PCA+CNN, CNN, and Suv regard MSE as the loss function, and CNN+Lss and Suv+Lss adopt the separable loss. Compared with PCA, CNN, and Suv, we can draw a conclusion that the features extracted by the self-supervised loss are more discriminative for image fusion. As can be seen in Table III, the Suv method has smaller spatial

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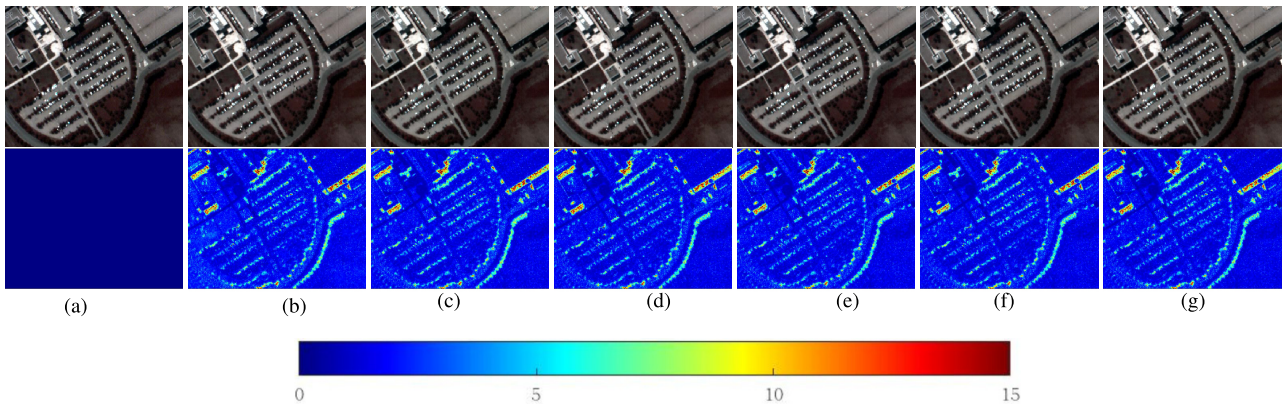

Fig. 4. False-color images (composed by bands 25, 15, and 10) are shown in the first and third rows, and SAM images are presented in the second and fourth rows. (a) Ground truth. (b) PCA. (c) PCA+CNN. (d) CNN. (e) Suv. (f) CNN+Lss. (g) Suv+Lss.

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT FUSION METHODS ON A SUBSCENE OF THE PAVIA UNIVERSITY DATASET

	PSNR	SAM	ERGAS	SSIM	RMSE	UIQI
Reference	∞	0	0	1	0	1
PCA	41.7425	2.2641	0.7549	0.9849	2.0865	0.9853
PCA+CNN	42.0566	2.1881	0.7366	0.9848	2.0124	0.9853
CNN	42.0226	2.1469	0.7193	0.9850	2.0203	0.9856
Suv	42.2432	2.1388	0.7213	0.9851	1.9696	0.9856
CNN+Lss	42.1312	**2.1253**	0.7299	0.9851	1.9952	0.9857
Suv+Lss	**42.3018**	2.1324	**0.7192**	**0.9852**	**1.9564**	**0.9858**

and spectral errors in terms of fusion results. For PCA, the transformed features lack valuable detail information for image reconstruction. There are more image details in the training set and test set, CNN can make the most of them. So we can get higher PSNR, SSIM, and UIQI values and lower spectral error than the traditional subspace methods. Compared with PCA, PCA+CNN extracts more structural information easily recognized by network. Whether spatial structure or spectrum, we can observe that PCA+CNN achieves better fusion effect closer to the reference. Besides, from the perspective of the spatial domain, CNN can extract more structure and detail information. In the restoration process of HSI, better spatial information of each band is restored, which also promotes spectral restoration to a certain extent. From the perspective of the spectrum, the rich features extracted by CNN can express the close relationship between spectral bands and further promote the recovery of the spectrum. For the network, the stronger the ability to extract the features that are most suitable for image fusion, the better a fusion task can be achieved. Based on this, Suv proposes extracting more discriminative features using the self-supervised loss technology. Compared with CNN, we can see clearly that the features learned by the self-supervised loss are more discriminative. Similarly, the same results can be obtained from the quantitative values of CNN+Lss and Suv+Lss. The results of CNN and CNN+Lss in Table III indicate that the loss can enhance image details, and Suv and Suv+Lss results demonstrate that it is effective in the spectrum. Of course, these results are related to the coefficient in Lss.

Fig. 4 shows the fusion results under different feature extraction methods and different loss functions. It can be seen from the subtle details of the image that Suv+Lss image is closest to the ground truth.

TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT FUSION METHODS ON A SUBSCENE OF THE CHIKUSEI DATASET

	PSNR	SAM	ERGAS	SSIM	RMSE	UIQI
Reference	∞	0	0	1	0	1
LTTR [49]	41.4927	2.9107	1.4659	0.9788	2.1474	0.9596
FUSE [44]	41.7510	2.0445	1.3218	0.9878	2.0844	0.9713
HySure [50]	46.0556	1.9094	1.2109	0.9917	1.2699	0.9743
LTMR [35]	43.7910	1.6321	1.7693	0.9923	1.6481	0.9756
OTD [51]	49.1794	1.3390	0.9440	0.9951	0.8853	0.9831
RSIFNN [37]	46.1286	1.8053	1.1699	0.9919	1.2592	0.9769
RSIFNN+Lss	46.5739	1.8403	1.1921	0.9917	1.1963	0.9777
Hyperpnn [16]	47.3615	1.5892	1.1071	0.9931	1.0926	0.9795
Hyperpnn+Lss	47.8595	1.5422	1.0517	0.9937	1.0317	0.9800
TFNet [52]	47.6119	1.5048	1.0071	0.9942	1.0616	0.9810
TFNet+Lss	48.0506	1.4758	1.0751	0.9937	1.0093	0.9813
(PDCon)SSF-CNN [32]	48.1443	1.4421	1.0558	0.9937	0.9984	0.9810
(PDCon)SSF-CNN+Lss	48.6808	1.3906	0.9731	0.9949	0.9386	0.9835
Suv+Lss	**49.2189**	**1.3065**	**0.9115**	**0.9954**	**0.8823**	**0.9846**

2) Experiments on Chikusei Dataset: The results of our method by comparing it with circumstances without self-supervised loss and separable loss show its effectiveness on the Chikusei dataset. The experiments are done based on a chosen area of the airborne image with high resolution.

Table IV shows the fusion results of different methods. The traditional image fusion methods LTTR, FUSE, HySure, LTMR, and OTD display better performance improvement in spatial details and spectral distortion compared with LR-HSI, especially OTD. With the stronger ability of CNN to extract valuable features for image fusion, the performance of fusion methods based on deep learning is gradually improved. Compared with the deep learning methods with the same number of convolution layers, our method Suv+Lss achieves the best performance due to self-supervised feature extraction and separable loss function. For the sake of proving the effectiveness of our Lss more fairly, we replace MSE with Lss based on the original deep learning methods, such as RSIFNN+Lss, Hyperpnn+Lss, TFNet+Lss, and (PDCon)SSF-CNN+Lss. Comparing these results, it can be concluded that Lss is effective.

Fig. 5 shows fused images, and there are relatively more red areas in SAM images based on the traditional fusion methods.

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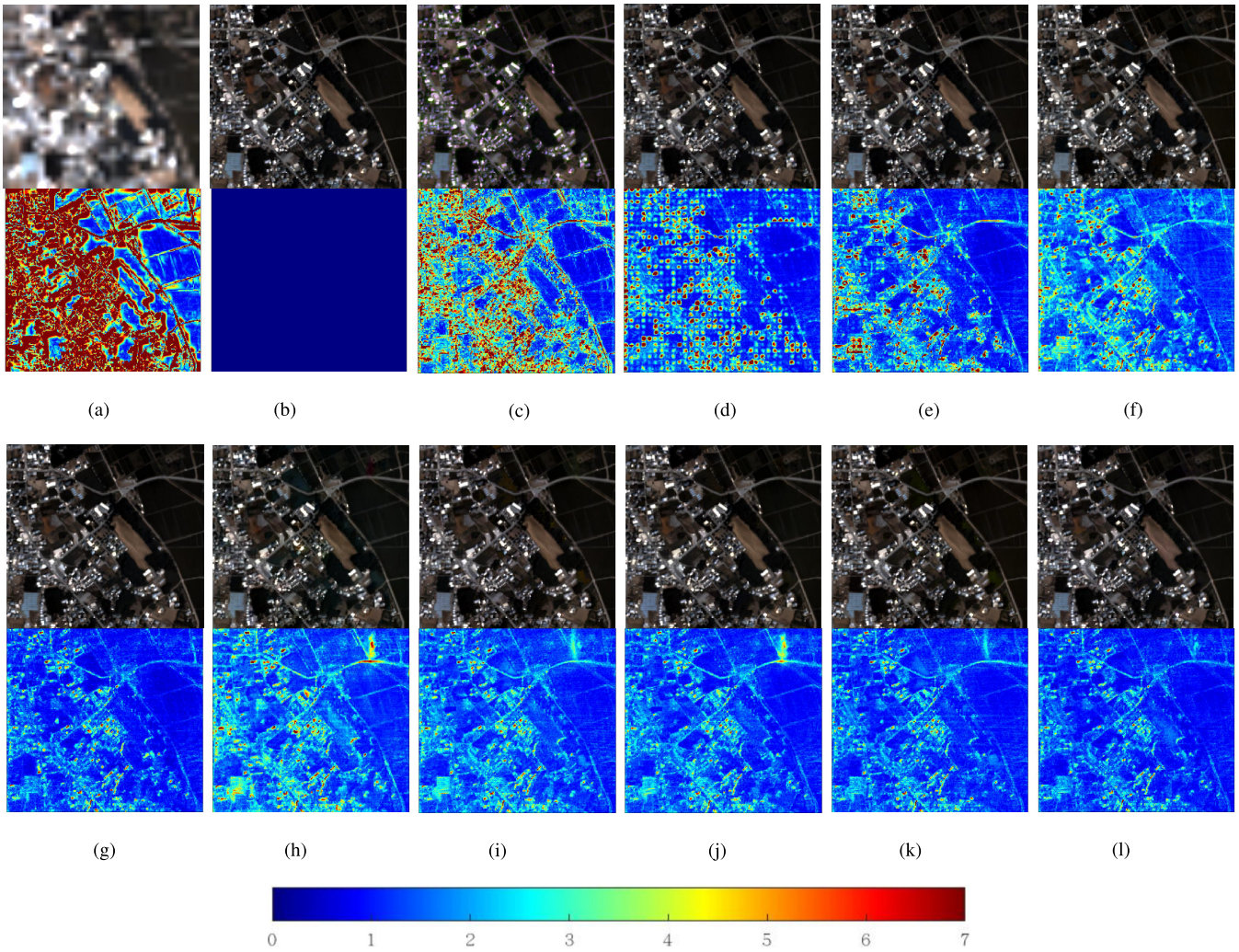

Fig. 5. False-color images (composed by bands 25, 15, and 10) are shown in the first and third rows, and SAM images are presented in the second and fourth rows. (a) LR-HSI. (b) Ground truth. (c) LTTR. (d) FUSE. (e) LTMR. (f) Hysure. (g) OTD. (h) RSIFNN. (i) Hyperpnn. (j) TFNet. (k) (PDCon)SSF-CNN. (l) Suv+Lss.

Under the same conditions, RSIFNN, Hyperpnn, and TFNet show more spectral distortion. Compared with (PDCon)SSF-CNN SAM image, our Suv+Lss SAM image has less cluttered details. Therefore, Suv+Lss can keep good spatial and spectral information.

As can be seen in Table V, PCA still keeps good spatial and spectral information due to more details provided by the data. From CNN and PCA+CNN, it can be noted that the features based on image structure are helpful to the restoration of images and they can greatly improve the performance of fusion. For Suv, the feature maps extracted by the self-supervised way are conducive to the learning of spatial details but it has little effect on the recovery of the spectrum, which is closely related to the coefficient of the self-supervised loss. By comparing the performance of CNN and CNN+Lss, as well as Suv and Suv+Lss, we can observe that the separable loss based on the spectrum and space effectively constrains the spatial and spectral information during training, which shows that our strategy improves the fusion effect.

Fig. 6 shows fused images. In SAM images, blue to red indicates that the SAM value increases gradually. All the methods can restore images to different degrees, and LTMR shows lower performance. It is obvious from the SAM diagram that the red area in the PCA SAM image is concentrated in the detail area of the image, where its spectrum distortion is serious. Compared with PCA, there are fewer red areas, and we can see from the upper right corner of these images that PCA+CNN and Suv+Lss show less spectral distortion. Compared with PCA+CNN, Suv+Lss image appears blue in more areas.

TABLE V
INDEX VALUES OF DIFFERENT FUSION METHODS ON A SUBSCENE OF THE CHIKUSEI DATASET

	PSNR	SAM	ERGAS	SSIM	RMSE	UIQI
Reference	∞	0	0	1	0	1
PCA	46.7875	1.6919	1.1283	0.9937	1.1672	0.9792
PCA+CNN	48.6871	1.3425	0.9517	0.9949	0.9380	0.9830
CNN	48.7093	1.3289	0.9213	0.9954	0.9356	0.9836
Suv	48.7177	1.3367	0.9763	0.9950	0.9347	0.9821
CNN+Lss	48.7799	1.3510	0.9489	0.9951	0.9280	0.9822
Suv+Lss	**49.2189**	**1.3065**	**0.9115**	**0.9954**	**0.8823**	**0.9846**

3) Experiments on Earth Observing-1 Hyperion and Sentinel-2A Dataset: The fusion results of the proposed

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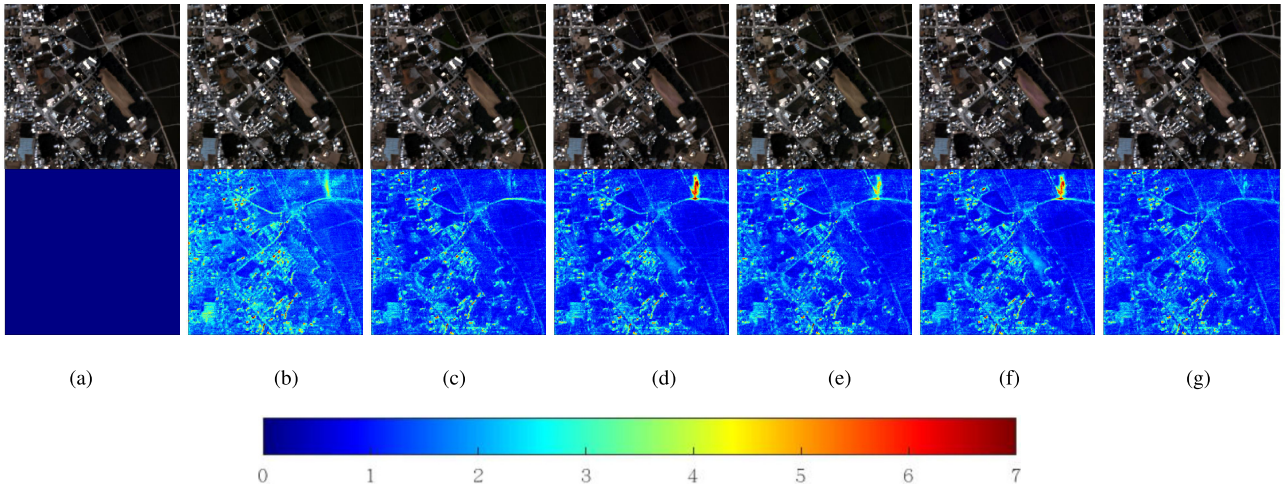

Fig. 6. False-color images (composed by bands 25, 15, and 10) are shown in the first and third rows, and SAM images are presented in the second and fourth rows. (a) Ground truth. (b) PCA. (c) PCA+CNN. (d) CNN. (e) Suv. (f) CNN+Lss. (g) Suv+Lss.

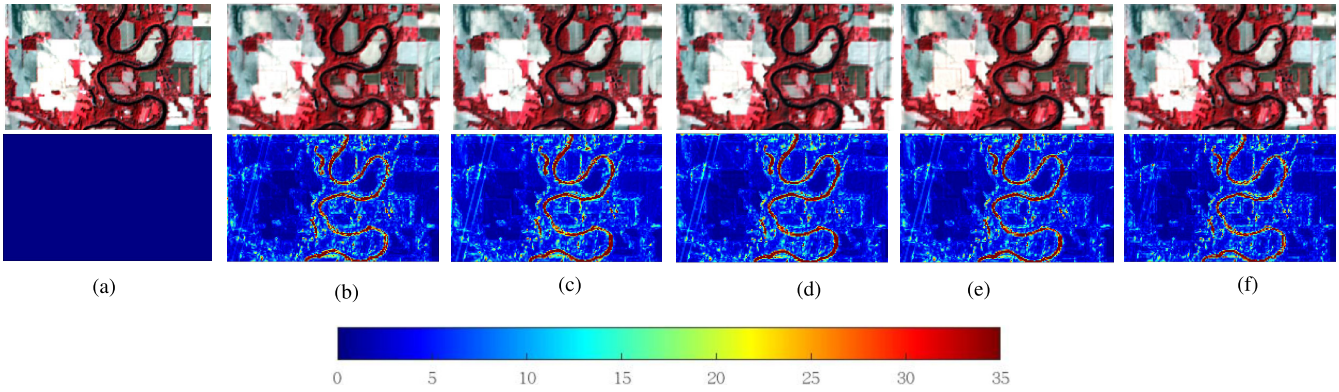

Fig. 7. False-color images (composed by bands 25, 15, and 10) are shown in the first and third rows, and SAM images are presented in the second and fourth rows. (a) Ground truth. (b) RSIFNN. (c) Hyperpnn. (d) TFNet. (e) (PDCon)SSF-CNN. (f) Suv+Lss.

technologies are compared with the methods without self-supervised loss and separable loss in this section.

Quantitative performance evaluation of fusion results about different methods is calculated based on the same conditions in Table VI. Compared with Suv+Lss, whether spatial structure or spectrum, we can observe that RSIFNN, Hyperpnn, TFNet, and (PDCon)SSF-CNN achieve lower performance. Besides, RSIFNN+Lss, Hyperpnn+Lss, TFNet+Lss, and (PDCon)SSF-CNN+Lss represent that the fusion models replaced MSE with the Lss loss function, and we can see that our Lss can extract more optimized features for subsequent image fusion from the results in Table VI.

Fig. 7 shows the false-color images and SAM images of the fused images captured by all the bands for different methods. From the fake-color image, we can see that Suv+Lss has more areas close to dark blue on the spectral image.

Quantitative performance evaluation of fusion results about different methods is carried out in Table VII, and PCA is farthest from the reference value. Compared with PCA, PCA+CNN extracts more features that are useful for restoring the image, and we can observe that PCA+CNN achieves performance closer to the reference. Besides, CNN has a better performance than PCA+CNN with the same dimensional features. For the network, the more valuable the extracted features are for image reconstruction, the better the fusion task can be achieved. Based on this, Suv extracts more discriminative features using the self-supervised feature extraction technology. To evaluate the performance of separable loss, CNN and CNN+Lss results in Table VII indicate that Lss can enhance image details, and Suv and Suv+Lss results demonstrate that it is effective in spectrum.

The false-color images and SAM images captured by all the bands for different methods are shown in Fig. 8. From the fake-color images, we can see the PCA image is less bright than other images, especially in the area around the

TABLE VI

PERFORMANCE COMPARISON OF DIFFERENT FUSION METHODS ON A SUBSCENE OF EARTH OBSERVING-1 HYPERION AND SENTINEL-2A DATASET

	PSNR	SAM	ERGAS	SSIM	RMSE	UIQI
Reference	∞	0	0	1	0	1
RSIFNN [37]	22.7177	6.9251	4.2885	0.8433	18.6490	0.9590
RSIFNN+Lss	22.7364	6.8597	4.2808	0.8478	18.6089	0.9599
Hyperpnn [16]	22.2669	7.3206	4.5245	0.8359	19.6425	0.9557
Hyperpnn+Lss	22.4061	7.1525	4.4529	0.8395	19.3301	0.9577
TFNet [52]	22.4243	7.0758	4.4308	0.8470	19.2897	0.9615
TFNet+Lss	22.6603	7.0492	4.3034	0.8525	18.7726	0.9621
(PDCon)SSF-CNN [32]	22.7538	7.0843	4.2693	0.8439	18.5717	0.9604
(PDCon)SSF-CNN+Lss	22.7992	7.0435	4.2498	0.8480	18.4747	0.9613
Suv+Lss	**23.1649**	**6.4595**	**4.0573**	**0.8528**	**17.7132**	**0.9618**

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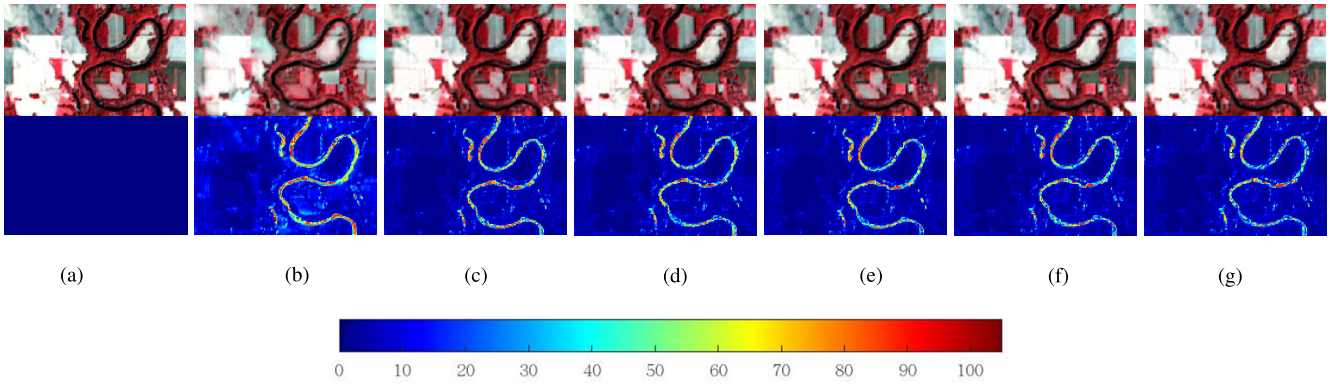

Fig. 8. False-color images (composed by bands 35, 25, and 10) are shown in the first and third rows, and SAM images are presented in the second and fourth rows. (a) Ground truth. (b) PCA. (c) PCA+CNN. (d) CNN. (e) Suv. (f) CNN+Lss. (g) Suv+Lss.

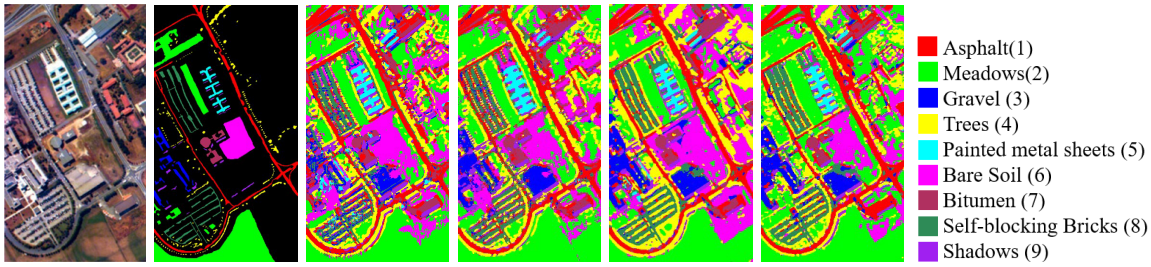

Fig. 9. Classification maps for the Pavia University dataset. From left to right, these images are the false-color composite image, ground truth, classification maps produced by the RSSAN based on images before and after fusion, classification maps produced by the A2S2K method based on images before and after fusion, and color labels.

TABLE VII

INDEX VALUES UNDER DIFFERENT FEATURE EXTRACTION METHODS AND LOSS FUNCTIONS ON EARTH OBSERVING-1 HYPERION AND SENTINEL-2A DATASET

	PSNR	SAM	ERGAS	SSIM	RMSE	UIQI
Reference	∞	0	0	1	0	1
PCA	17.0589	11.0088	8.3943	0.6787	35.7761	0.8421
PCA+CNN	23.0339	6.8654	4.1163	0.8475	17.9823	0.9604
CNN	23.0870	6.6652	4.0934	0.8525	17.8727	0.9611
Suv	23.1041	6.6293	4.0810	0.8511	17.8376	0.9613
CNN+Lss	23.1117	6.5954	4.0749	**0.8549**	17.8219	0.9618
Suv+Lss	**23.1649**	**6.4595**	**4.0573**	0.8528	**17.7132**	**0.9618**

river. In SAM images, more areas are away from dark blue, compared with PCA+CNN, CNN, Suv, CNN+Lss, and Suv+Lss SAM images. From the results, it can be seen that the feature extraction method of PCA does not consider the subsequent fusion effect, and only processes the current data. Therefore, the network by learning can not only maintain PCA information but also extract discriminative features for the purpose of achieving good fusion effect. There is the smallest difference between Suv+Lss and ground truth, and more areas on the Suv+Lss spectral image are close to blue. Suv and Lss can further improve the local spectral and spatial performance compared with the existing methods.

C. Running Time

All the experiments of the deep-learning-based fusion models are implemented on GeForce GTX 1080 graphics processing unit (GPU) with 32-GB memory. Table VIII shows the test time (s) of the fusion models on the Pavia University, Chikusei, and Hyperion datasets, respectively. In all the compared methods, the proposed Suv+Lss lacks time advantage, mainly due to RDB with a large number of parameters.

TABLE VIII

TEST TIME (S) FOR DIFFERENT FUSION MODELS

Datasets	Methods				
	RSIFNN	Hyperpnn	TFNet	(PDCon)SSF-CNN	Suv+Lss
Pavia University	0.21	0.19	0.49	0.17	0.29
Chikusei dataset	0.22	0.18	0.41	0.19	0.33
Hyperion dataset	0.25	0.26	0.38	0.20	0.31

D. Land-Cover Classification

Land-cover classification is one of the important applications of HSI. To verify the practicability of the proposed model, our fusion method is tested in experiments about land-cover classification conducted on the Pavia University dataset. In this article, the data are sampled twice using the same spatial kernel. The first sampled Pavia University data and the second sampled Pavia University data are used as training pairs, and then the first sampled image is used for testing. The fused Pavia University data are for land-cover classification. Three accuracy metrics, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC), are calculated to measure the performance. Among them, OA is the ratio of the correctly classified samples to the total test samples, AA represents the average of classwise accuracy, and KC measures a strong mutual agreement between the predicted classification maps and the ground truth. The classification results on our fused images are shown based on the representative deep learning methods A2S2K [54] and RSSAN [55].

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TABLE IX
OA, AA, AND KC VALUES ON THE PAVIA UNIVERSITY DATASET

Class	1	2	3	4	5	6	7	8	9	OA	AA	KC
A2S2K_un	90.27	97.77	91.93	97.11	99.54	90.37	96.52	93.03	99.83	95.06	95.15	93.45
A2S2K [54]	95.66	99.28	97.88	94.81	99.92	93.23	94.20	94.92	99.78	97.12	96.63	96.17
RSSAN_un	77.09	93.64	62.05	94.01	98.28	74.79	86.95	74.25	98.34	85.70	84.37	82.07
RSSAN [55]	91.53	99.19	80.66	95.37	100.00	90.26	94.12	91.32	99.78	94.98	93.58	93.32

As shown in Table IX, we give the objective classification results based on the images before and after fusion, where A2S2K_un and RSSAN_un represent the classification results of A2S2K and RSSAN based on not fused images, respectively, A2S2K and RSSAN stand for the classification results of A2S2K and RSSAN, respectively, based on our fused images, and 1–9 indicates nine categories in Fig. 9. Obviously, the image fusion strategy can improve the AA of classification, but it can also reduce the classification accuracy of a category in rare cases, such as Trees in A2S2K. According to the classification results of the following two methods, if the classification method itself cannot extract features to fully represent the target, the classification method based on fusion is more conducive to the improvement of classification accuracy. At the same time, we also show the classification maps based on the images before and after fusion. From the figures, it can be seen that the category boundary is more clearly distinguished on the fused images.

V. CONCLUSION

In this study, to overcome these existing problems of HSI and MSI fusion technologies based on deep learning, such as samples with poor discrimination, sample imbalance spatial domain and spectral domain, and pixel constraints for discarding the physical meaning of HSI, we present a fusion method containing two strategies about generating discriminative samples based on Suv and Lss. For obtaining discriminative samples and balancing spatial and spectral samples, we introduce a self-supervised scheme based on subspace. First, labels are generated using the traditional subspace technology, and then low-dimensional features are extracted including almost all the original image information based on Suv. After that, we merge with the structural features extracted by the end-to-end network to obtain the most valuable features for image restoration. To make full use of the physical characteristics of HSI in training, the loss Lss is constructed spatially and spectrally by further understanding the 3-D characteristic of HSIs. Extensive experiments are implemented on three public benchmark datasets, which has shown improvements in our fusion method.

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